

1. In a group of 50 people, 28 like tea, 26 like coffee, and 12 like both. Find the number of people who like only coffee.
2. In a group of 60 people, 27 like cold drinks and 42 like hot drinks and each person likes at least one of the two drinks. How many like both coffee and tea?
3. In a group of 50 people, 28 have travelled to Europe, 31 have travelled to Asia, and 10 have travelled to both continents. How many people have not travelled to either continent?
4. In a survey of 150 people, 95 like apples, 70 like bananas, and 60 like both apples and bananas. Find the number of people who like only apples.
5. A survey of 200 people found that 120 read newspaper A, 150 read newspaper B, and 90 read both. How many read at least one of the newspapers?
6. In a group of 60 people, 27 like cold drinks and 42 like hot drinks and each person likes at least one of the two drinks. How many like both coffee and tea?
7. In a class of 50 students, each student passed either in mathematics or science or in both. 10 students passed in both and 28 passed in science. Find how many students passed in mathematics?
8. In a class of 212 students, each student studies at least one of the three subjects Mandarin, French and Spanish. 96 of them study Mandarin, 102 studies French and 106 Spanish. 32 studies Mandarin and French, 34 study Mandarin and Spanish and 36 study French and Spanish. How many students study all the three subjects?
9. In a sample of 100 logic chips, 23 have a defect D1, 26 have a defect D2, 30 have a defect D3, 7 have defects D1 and D2, 8 have defects D1 and D3, 10 have defects D2 and D3 and 3 have all the three defects. Find the number of chips having
 - (i) Atleast one defect
 - (ii) No defect

10. A survey of a sample of 25 new cars being sold by an auto dealer was conducted to see which of the three popular options; air conditioning, radio and power windows were installed. The survey found that 15 had air-conditioning, 12 had radio, 11 had power windows, 5 had AC and PW, 9 had AC and radio, 4 had radio and PW and 3 had all the three options. Find the number of cars that has
- (i) Only power windows
 - (ii) Only air-conditioning
 - (iii) Only radio
 - (iv) Only one of the options
 - (v) At least one option
 - (vi) None of the options
11. Thirty cars are assembled in a factory. The options available are a music system, an air conditioner and power windows. It is known that 15 of the cars have music systems, 8 have air conditioners and 6 have power windows. Further 3 have all the options. Determine at least how many cars do not have any option at all?
12. Among the students in a hostel, 12 students study Mathematics (A), 20 study Physics (B), 20 study Chemistry (C) and 8 study Biology (D). There are 5 students for A and B, 7 students for A and C, 4 students for A and D, 16 students for B and C, 4 students for B and D and 3 students for A, C and D. Finally, there are 2 students who study all these subjects. Furthermore, there are 71 students who do not study any of these subjects. Find the total number of students in the hostel.